

Applied Mathematics

Most Frequently Asked & Important Questions

IT 4th Semester • Pokhara University • BE Programme

Papers Analyzed

Spring 2024 | Fall 2024 | Spring 2025 | Fall 2025

This guide analyzes all four available Pokhara University Applied Mathematics question papers (4th semester, BE) topic-by-topic to show exactly which questions repeat, how often, and in what form — so you can prioritize revision time on what actually gets asked.

Priority	Meaning	Approx. Weight
Must Do	Appears in all 4 papers, often nearly word-for-word	~55 marks/paper
High Priority	Appears in 3 of 4 papers	~15 marks/paper
Good to Know	Appears in 2 of 4 papers / short questions (Q7)	~10 marks/paper

How the marks break down (each paper is 100 marks, 3 hrs):

Unit 1 – Complex Analysis (Analyticity, Cauchy formulas, Laurent series, bilinear transform) $\approx$ 30 marks |
Unit 2 – Z-Transform & Difference Equations $\approx$ 15 marks | Unit 3 – PDEs (Wave, Heat, Laplace) $\approx$ 23 marks
| Unit 4 – Fourier Transform/Integral $\approx$ 15 marks | Short Q7 $\approx$ 10 marks

UNIT 1 — Complex Analysis

ASKED IN ALL 4 PAPERS — MUST DO

Define analyticity of $f(z) = u(x,y) + iv(x,y)$. Prove/derive the Cauchy-Riemann (C-R) equations $u_x = v_y$, $u_y = -v_x$ as necessary conditions for analyticity.

Seen in: Sp'24 Q1a, Fa'24 Q1a, Sp'25 Q1a, Fa'25 Q1a • Typical marks: 5–7

→ This is the opening question every single time. Know the full proof (Cauchy-Riemann from $f'(z)$ along real and imaginary directions), not just the statement.

ASKED IN ALL 4 PAPERS — MUST DO

State and prove Cauchy's Integral Formula (or the derivative form). Evaluate a contour integral of the form $\oint_C e^{kz}/(z-a)^n dz$ using it (e.g. $e^{5z}/(z+i)^4$, $e^{2z}/(z-1)^4$, $z^6/(2z-1)^6$).

Seen in: Sp'24 Q1b, Fa'24 Q1b, Sp'25 Q2b, Fa'25 Q1c • Typical marks: 5–8

→ Same skeleton every time: state formula → prove it → apply for n -th derivative case. Practice differentiating e^{kz} up to 3rd order fast.

ASKED IN ALL 4 PAPERS — MUST DO

Cauchy's Residue Theorem: state it (and sometimes prove it), then evaluate a contour integral with multiple poles using residues, e.g. $\oint 2z / [(z+1)(z-1)^3(z+3)] dz$, $\oint dz/(z^2+4)^3$, $\oint 2z/[(z-1)(z+3)] dz$.

Seen in: Sp'24 Q1-OR, Fa'24 Q2a, Sp'25 Q7a, Fa'25 (via Q2b) • Typical marks: 5–8

→ Always appears either as a main question OR inside the short-answer set. Practice residues at simple poles AND higher-order poles (order-3 pole in Sp'24/Sp'25).

ASKED IN ALL 4 PAPERS — MUST DO

Laurent series expansion of a rational function in 2–3 given annular regions, e.g. $(7z-2)/[z(z+1)(z-2)]$, $(4z+3)/(z^3-z^2-6z)$, $1/(z^2-3z+2)$, $(z+3)/[z(z+1)(z-2)]$ for $1 < |z| < 2$.

Seen in: Sp'24 Q2a, Fa'24 Q7a, Sp'25 Q2a, Fa'25 Q2a • Typical marks: 5–7

→ Do partial fractions first, then expand each term as a geometric series valid in the given $|z|$ range. Practice all three region types: $|z| < r$, $r_1 < |z| < r_2$, $|z| > r$.

ASKED IN ALL 4 PAPERS — MUST DO

Define bilinear (Möbius) transformation. Find the transformation mapping $z = 0, -1, i$ onto $w = i, 0, \infty$ (this exact mapping repeats). Find the image of the unit circle $|z| = 1$ under it.

Seen in: Sp'24 Q2b, Fa'24 Q2b (different points), Sp'25 Q2-OR, Fa'25 Q2c • Typical marks: 5–8

→ The points $z=0, -1, i \rightarrow w=i, 0, \infty$ appear identically in 3 of the 4 papers — memorize this exact worked solution. Also practice the general cross-ratio method in case different points are given (as in Fa'24: $0, 1, \infty \rightarrow -3, -1, 1$).

Also seen once/twice (know the method, lower priority):

- Harmonic functions & harmonic conjugate ($u = \sin x \cdot \cosh y$, $y^3 - 3x^2y$) — Sp'25 Q1b, Fa'25 Q1b, Fa'24 Q7c
- Singularity/zeros/poles definitions — Fa'25 Q2b
- Short Q7: check analyticity of z^2 or $z \bar{z}$; solve $u_x + u_y = 0$ by separation of variables

UNIT 2 — Z-Transform & Difference Equations

ASKED IN ALL 4 PAPERS — MUST DO

State and prove the first shifting theorem on Z-transform. Find $Z(e^{in\pi/2})$ and hence deduce $Z(\cos n\pi/2)$ and $Z(\sin n\pi/2)$. (Sometimes phrased as $Z[a^n f(t)] = F(z/a)$ with the same e^{ibt} application.)

Seen in: Sp'24 Q3a, Fa'24 Q3a, Sp'25 Q3a&b, Fa'25 Q3a • Typical marks: 7–8

→ Word-for-word identical in Sp'24 and Fa'24. This is close to a guaranteed 7-8 marks — memorize the derivation using the geometric series definition of Z-transform.

ASKED IN ALL 4 PAPERS — MUST DO

Solve a linear difference equation using Z-transform, given y_0, y_1 (e.g. $y_{n+2} - 7y_{n+1} + 12y_n = 2n$; $y_{n+2} + 6y_{n+1} + 9y_n = 2^n$; $U_{n+2} - 2\cos\alpha U_{n+1} + U_n = 0$).

Seen in: Sp'24 Q3b, Fa'24 Q3b, Sp'25 Q4a, Fa'25 Q3b • Typical marks: 7–8

→ Practice the full workflow: take Z-transform of both sides → apply shifting theorems → solve for $Y(z)$ algebraically → partial fractions → inverse Z-transform.

ASKED 3 OF 4 PAPERS — HIGH PRIORITY

Show $Z[nf(t)] = -z \frac{d}{dz}[F(z)]$. Find inverse Z-transforms such as $Z^{-1}[z/((z+1)^2(z-1))]$, $Z^{-1}[z/(z^2-5z+6)]$, $Z^{-1}[z/((z+2)(z-1)^2)]$.

Seen in: Sp'24 Q4a, Fa'24 Q4a, Sp'25 Q3a(i,ii) • Typical marks: 3–7

→ Partial fraction decomposition is the key skill here — practice repeated-root cases like $(z-1)^2$ carefully.

Also worth knowing: $Z(a^n)$, $Z(n^2)$, $Z(a^n \cos bt)$ appear as short (2.5–5 mark) questions almost every paper.

UNIT 3 — Partial Differential Equations (Wave, Heat, Laplace)

ASKED IN ALL 4 PAPERS — MUST DO

Derive the one-dimensional wave equation OR one-dimensional heat equation from first principles with all necessary physical assumptions.

Seen in: Fa'24 Q4b, Sp'25 Q4b-OR, Fa'25 Q3b(2025-Spring style), Sp'24 (heat eq context) • Typical marks: 7–8

→ Derivations rotate between wave and heat equation each sitting — prepare BOTH derivations fully; one of them shows up every time.

ASKED IN ALL 4 PAPERS — MUST DO

Solve the 1D heat equation $\partial u / \partial t = c^2 \partial^2 u / \partial x^2$ for a bar of length L with ends at 0° , given a piecewise initial temperature $f(x)$ (triangular / tent-shaped, e.g. $f(x)=x$ for $0 < x < L/2$, $f(x)=L-x$ for $L/2 < x < L$).

Seen in: Sp'24 Q5b, Fa'24 Q5-OR, Sp'25 Q5a&b, Fa'25 Q5b • Typical marks: 7–8

→ The exact 'tent function' boundary condition (x then $L-x$, or symmetric variants) repeats in nearly every paper — know the Fourier half-range sine series set-up cold, including the b_n integral by parts.

ASKED 3 OF 4 PAPERS — HIGH PRIORITY

Express the Laplacian $\nabla^2 u$ in polar coordinates (derive fully) — often offered as an OR alternative to deriving the 2D heat equation or the 1D wave equation.

Seen in: Sp'24 Q5a, Fa'24 Q5a, Sp'25 Q4b-OR • Typical marks: 7–8

→ Standard substitution $x=r \cos \theta$, $y=r \sin \theta$ and chain rule — long but mechanical; practice it once and you can reproduce it under time pressure.

ASKED 2 OF 4 PAPERS

Solve the 1D wave equation with a given initial deflection and zero (or given) initial velocity (e.g. deflection $= 0.01 \sin 3x$, $L=\pi$, $c^2=1$), or find the deflection of a square membrane $u(x,y,t)$ with $a=b=1$ and initial deflection like $0.1 \sin 3\pi x \cdot \sin 4\pi y$.

Seen in: Sp'24 Q4b, Fa'25 Q5a, Fa'25 Q6b • Typical marks: 7–8

→ When the initial deflection is already a single sine term, the answer collapses to one term of the double/single Fourier series — don't do a full Fourier expansion, just match coefficients directly.

Also seen: 2D Laplace/steady-state problems on rectangular or semicircular plates (Fa'24 Q5b, Fa'25 Q6a); derive 2D heat equation (Sp'25 Q5-OR); short Q7 — solve $u_{xx} + 9u = 0$ or $u_x + u_y = 0$ by separation of variables.

UNIT 4 — Fourier Transform & Fourier Integral

ASKED IN ALL 4 PAPERS — MUST DO

Using Fourier integral representation, show the classic piecewise result: $\int_0^\infty (\cos xw + w \sin xw)/(1+w^2) dw = 0$ if $x < 0$, $\pi/2$ if $x=0$, πe^{-x} if $x > 0$.

Seen in: Sp'24 Q6b, Fa'24 Q6b, Sp'25 Q6a, Fa'25 Q4a • Typical marks: 7–8

→ Appears essentially word-for-word in all four papers — this is the single most reliable 7-8 marks in the whole exam. Memorize the derivation via Fourier integral theorem applied to e^{-x} .

ASKED IN ALL 4 PAPERS — MUST DO

Find the Fourier cosine transform of $f(x) = e^{-mx}$ ($m > 0$), then use it to show $\int_0^\infty \cos kx/(1+x^2) dx = (\pi/2)e^{-k}$.

Seen in: Sp'24 Q6a, Fa'24 Q6-OR, Sp'25 Q4b, Fa'25 Q4b • Typical marks: 7–8

→ Pairs with the question above — together these two make up an almost-guaranteed full sub-question in every paper. Also be ready for the variant using Parseval's identity to prove $\int dx/(1+x^2)^2 = \pi/4$ (Fa'25).

Also seen once: Fourier transform of xe^{-x^2} (Fa'24 Q6-OR); show $\int w \sin xw/(a^2+w^2) dw = (\pi/2)e^{-ax}$ (Fa'25 Q7b).

Quick-Revision Priority List

If you only have limited time before the exam, drill these in order — each one has shown up in **all four** available papers:

- 1. C-R equations: definition + full proof of analyticity condition
- 2. Cauchy Integral Formula: statement, proof, and derivative-form evaluation ($e^{kz}/(z-a)^n$)
- 3. Cauchy Residue Theorem: statement + evaluating contour integrals with multiple/higher-order poles
- 4. Laurent series expansion of a rational function in multiple annular regions
- 5. Bilinear transformation mapping $z = 0, -1, i$ onto $w = i, 0, \infty$, plus image of $|z| = 1$
- 6. Z-transform first shifting theorem + $Z(e^{in\pi/2})$, $Z(\cos n\pi/2)$, $Z(\sin n\pi/2)$
- 7. Solving a 2nd-order linear difference equation via Z-transform
- 8. 1D heat equation solution for a bar with tent-shaped (triangular) initial temperature
- 9. Deriving the 1D wave equation AND the 1D heat equation (both, from assumptions)
- 10. Fourier integral representation proof of the standard piecewise result $(\cos xw + w \sin xw)/(1+w^2)$
- 11. Fourier cosine transform of e^{-mx} and the resulting $\int \cos kx/(1+x^2) dx$ identity

Note: This analysis is based on pattern-matching across 4 available past papers (Spring 2024, Fall 2024, Spring 2025, Fall 2025) for the Pokhara University BE Applied Mathematics course. Past-paper frequency is a strong guide but not a guarantee of what will be asked — use this to prioritize study order, not to skip topics entirely.